MILITARY SPECIFICATION

SIGHTUNIT: M53A1

1. SCOPE

1.1 This specification covers a fire control instrument which is composed of Elbow Telescope: M109 and Telescope Mount: M128A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-T-46324	Telescope, Elbow: M109
MIL-M-50380	Mount, Telescope: M128A1

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army Frankford Arsenal

F8588780	Telescope, Elbow: M109
F10559698	Sightunit: M53A1
F10559699	Mount, Telescope: M128A1

PACKAGING DATA SHEET

10559698	Packaging of Sightunit: M53A1

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(Copies of specifications, standards, drawings, and packaging data sheet required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The sightunit shall be manufactured in accordance with Drawing F10559698 and drawings pertaining thereto.

3.1.1 Telescope.- The telescope shall be fabricated in accordance with Drawing F8588780 and drawings pertaining thereto, and shall meet the requirements of MIL-T-46324.

3.1.2 Mount.- The telescope mount shall be fabricated in accordance with Drawing F10559699 and drawings pertaining thereto, and shall meet the requirements of MIL-M-50380.

3.2 General specification.- The contractor shall be responsible for adherence to, and compliance with, the requirements of MIL-F-13926 listed below:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.3 Telescope retention.- The telescope clamping mechanism shall prevent movement of the telescope within the telescope mount when a 40 pound-inch torque is applied to the telescope body. This requirement shall be met at an ambient temperature between +60° and +90° Fahrenheit (F).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three sightunits

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manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I and II herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I

CLASSIFICATION CRITICAL: NONE MAJOR: AQL 1.0% Defective	REQUIREMENT	TEST PROCEDURE
101. Telescope retention	3.3	4.6.1
MINOR: NONE		

NOTE: The test in Table I shall be conducted at a standard ambient temperature of between 60° and 90°F.

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4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- Three sightunits shall be selected at random as a special sampling from each 100 produced. The samples shall meet the requirements and tests in Table II.

TABLE II

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
301.	Fabrication	3.1, 3.1.1, & 3.1.2	Applicable drawings and specifications - Visual
302.	General specifications	3.2	MIL-F-13926 - Visual

NOTE: The examinations in Table II shall be conducted at a standard ambient temperature of between 60° and 90°F.

4.4.2 Failure of sample.- Should any item of a special sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- When the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the Technical Data Package (TDP).

4.5.2.2 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of

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test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Telescope retention.- The telescope clamping mechanism shall be tested in accordance with standard shop practices using a commercial torque wrench accurate to within 1/4 pound inch. An appropriate adapter shall be utilized to permit application of the torque wrench to the telescope body. The telescope shall show no movement when the specified torque is applied to the telescope body in accordance with 3.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10559698. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The sightunit is used as an indirect fire control instrument for laying mortars and rocket launchers in elevation and azimuth.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging, and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet number (see 5.1).
(e) Provisions for First Article testing.

Custodian:

Army - MU

Preparing activity:

Army - MU

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